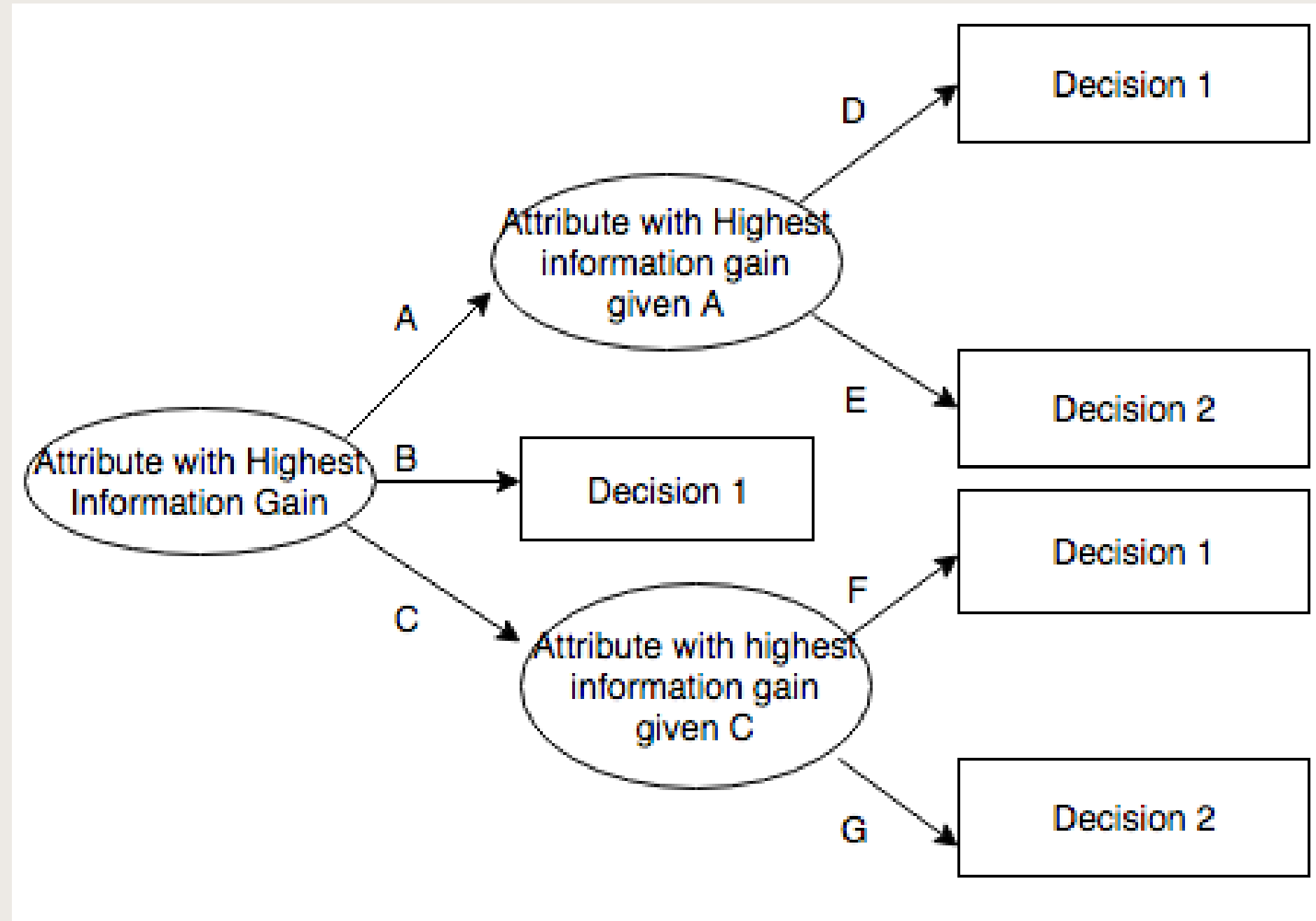


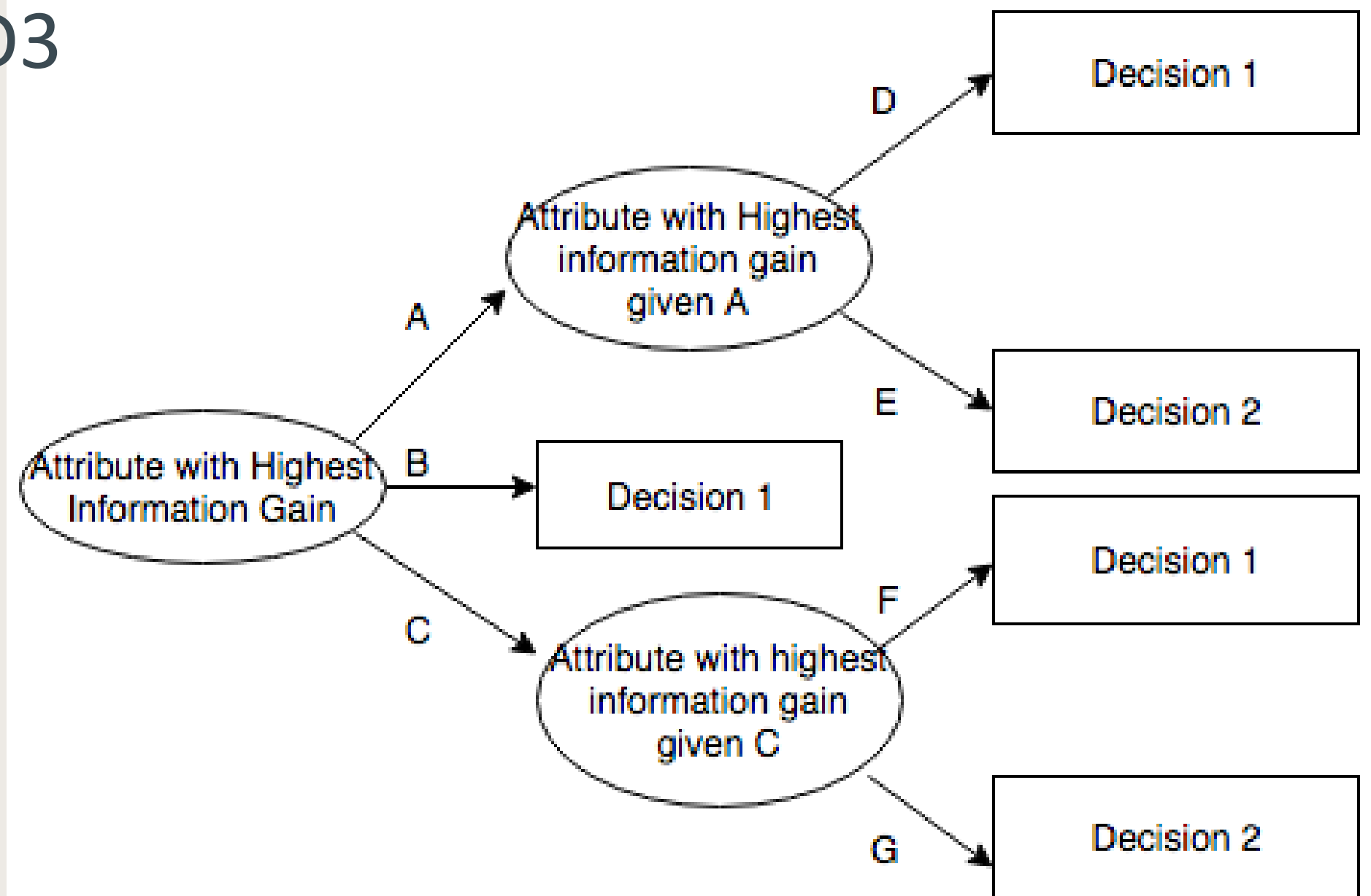
WHAT IS ID3

- ID3 stands for Iterative Dichotomizer3 and is named such because the algorithm iteratively(repeatedly) dichotomizes(divides) features into two or more groups at each step. ID3 is an algorithm invented by Ross Quinlan used to generate a decision tree from a dataset and is the most popular algorithms used to constructing trees.
- ID3 is the core algorithm for building a decision tree. It employs a top-down greedy search through the space of all possible branches with no backtracking. This algorithm uses information gain and entropy to construct a classification decision tree.



MAJOR CHARACTERISTICS OF THE ID3 ALGORITHM

- ID3 can overfit the training data (to avoid overfitting, smaller decision trees should be preferred over larger ones).
- This algorithm usually produces small trees, but it does not always produce the smallest possible tree.



Advantage and Disadvantage

ADVANTAGE

- Inexpensive to construct
- Extremely fast at classifying unknown records Easy to interpret for small-sized trees.
- Robust to noise (especially when methods to avoid over-fitting are employed).
- Can easily handle redundant or irrelevant attributes (unless the attributes are interacting).

DISADVANTAGE

- The space of possible decision trees is exponentially large. Greedy approaches are often unable to find the best tree.
- Does not take into account interactions between attributes.
- Each decision boundary involves only a single attribute.



SIMPLE ID3 EXAMPLE:

Weather	Temperature	Humidity	Windy	Play Tennis?
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

1

Calculating Entropy

- Positive instances (Play Tennis = Yes): 9
- Negative instances (Play Tennis = No): 5
- Entropy(S) = $-(9/14) * \log_2(9/14) - (5/14) * \log_2(5/14) \approx 0.940$

2

Calculating Information Gain

- $\text{Information Gain}(S, \text{Weather}) = \text{Entropy}(S) - [(5/14) * \text{Entropy}(\text{Sunny}) + (4/14) * \text{Entropy}(\text{Overcast}) + (5/14) * \text{Entropy}(\text{Rainy})] \approx 0.246$
- $\text{Information Gain}(S, \text{Temperature}) = \text{Entropy}(S) - [(4/14) * \text{Entropy}(\text{Hot}) + (4/14) * \text{Entropy}(\text{Mild}) + (6/14) * \text{Entropy}(\text{Cool})] \approx 0.029$
- $\text{Information Gain}(S, \text{Humidity}) = \text{Entropy}(S) - [(7/14) * \text{Entropy}(\text{High}) + (7/14) * \text{Entropy}(\text{Normal})] \approx 0.152$
- $\text{Information Gain}(S, \text{Windy}) = \text{Entropy}(S) - [(8/14) * \text{Entropy}(\text{False}) + (6/14) * \text{Entropy}(\text{True})] \approx 0.048$

3

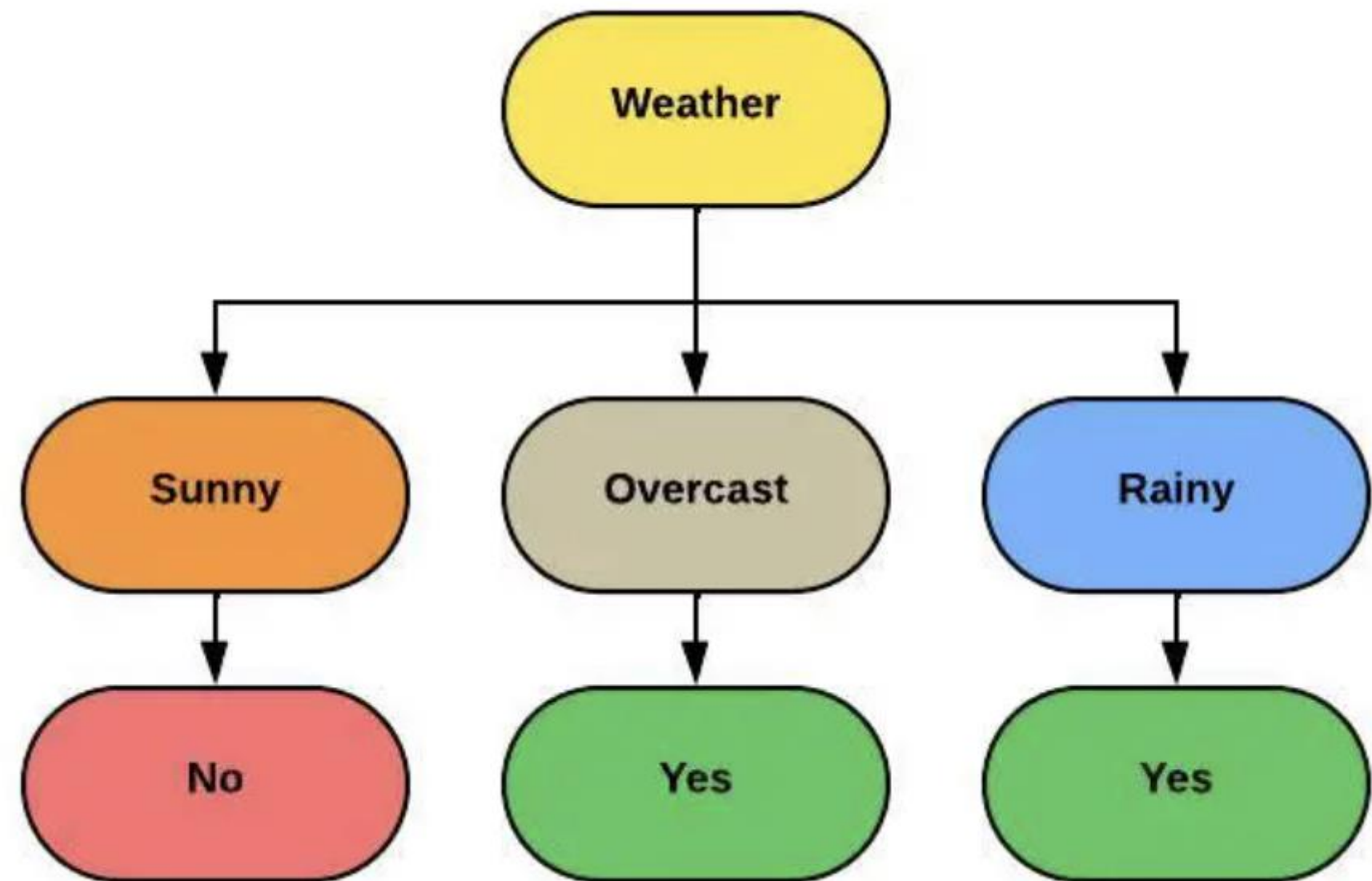
Selecting the Best
Attribute

4

Splitting the
Dataset

5

Repeat the Process





SIMPLE ID3 EXAMPLE:

Weather	Temperature	Humidity	Windy	Play Tennis?
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

5

Repeat the Process

SUNNY:

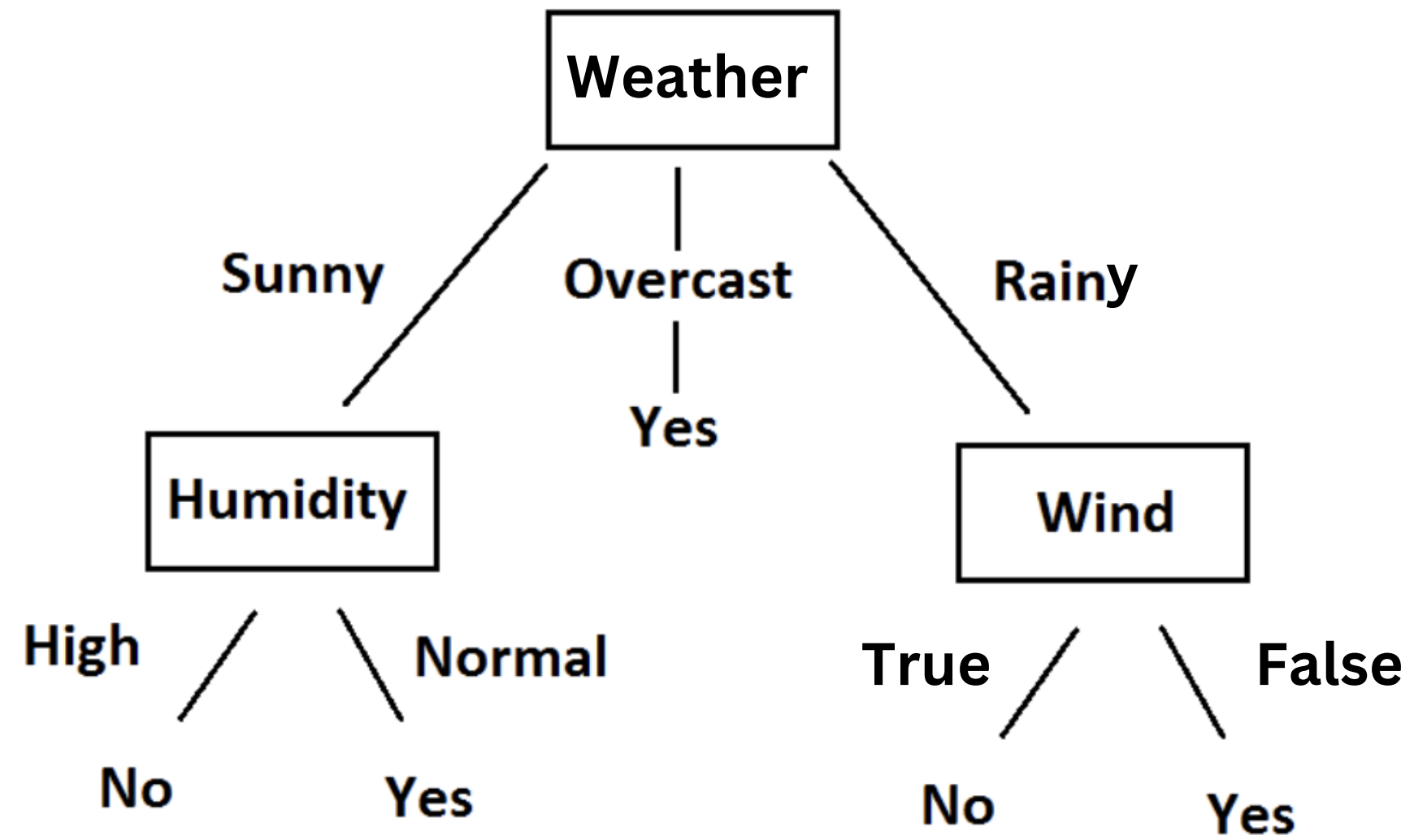
- $\text{Gain}(\text{Outlook}=\text{Sunny Temperature}) = 0.570$
- $\text{Gain}(\text{Outlook}=\text{Sunny Humidity}) = 0.970$
- $\text{Gain}(\text{Outlook}=\text{Sunny Wind}) = 0.019$

RAINY:

- $\text{Gain}(\text{Outlook}=\text{Rain Temperature}) = 0.01997309402197489$
- $\text{Gain}(\text{Outlook}=\text{Rain Humidity}) = 0.01997309402197489$
- $\text{Gain}(\text{Outlook}=\text{Rain Wind}) = 0.9709505944546686$

6

Final Result



ENTROPY AND GAIN



ENTROPY

Entropy is an information theory metric that measures the impurity or uncertainty in a group of observations(dataset). It determines how a decision tree chooses to split data.

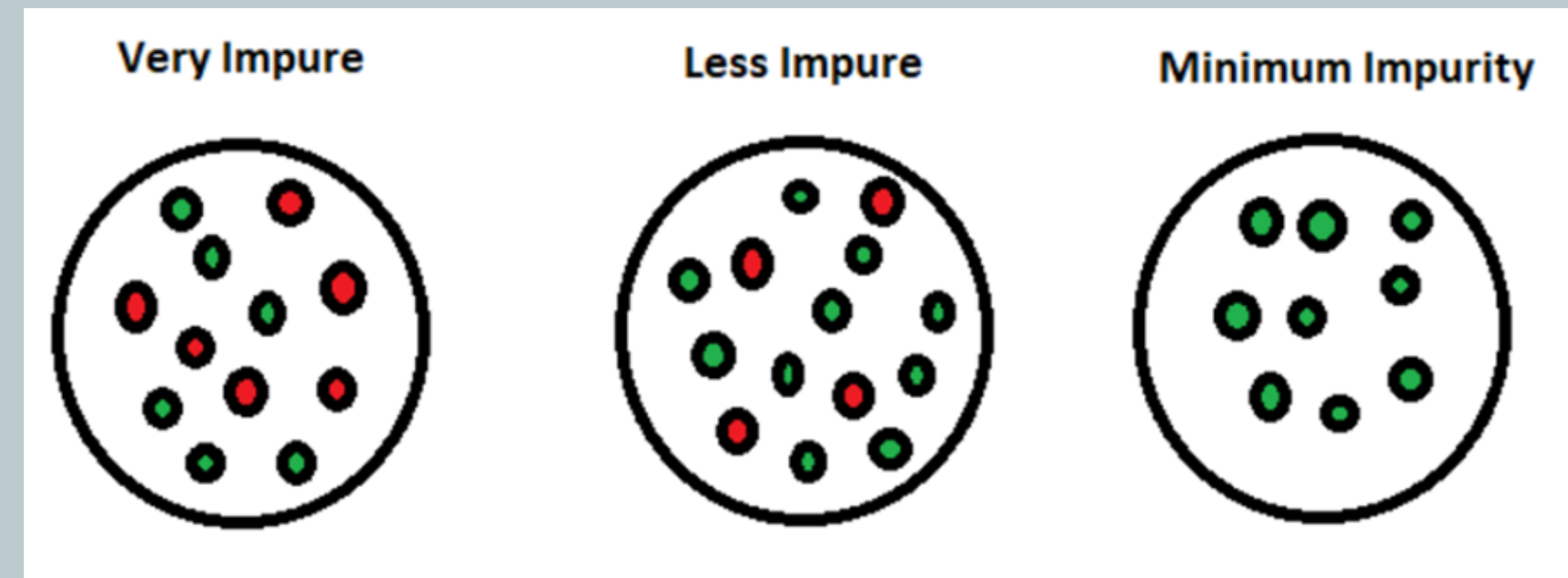
low = pure & predictable

high = mixed & unpredictable

FORMULA :

$$Entropy = \sum_{i=1}^C -p_i * \log_2(p_i)$$

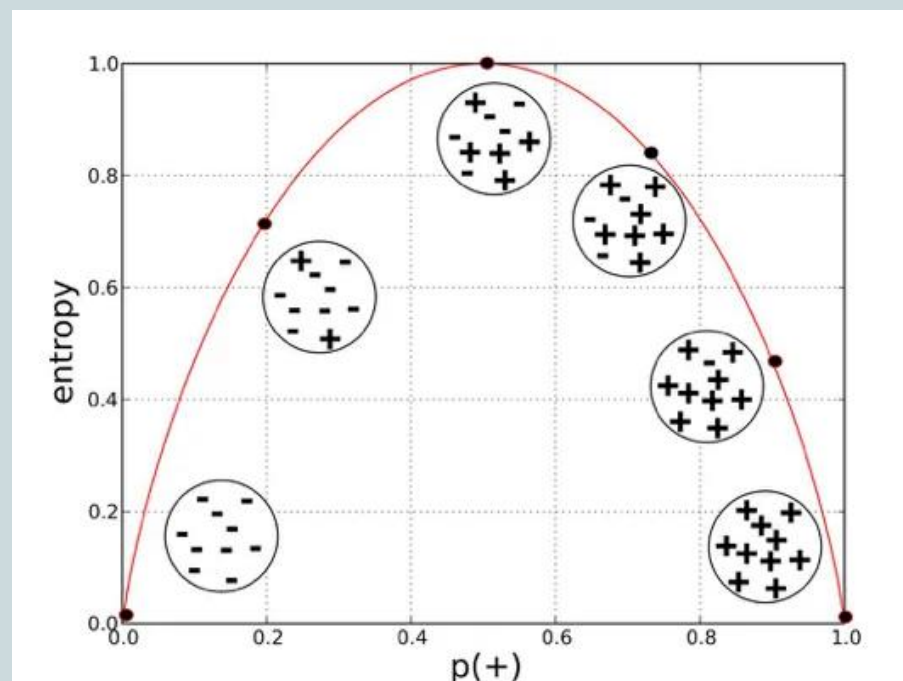
p_i is the proportion of instances in class i .



When entropy is zero the sample is completely homogeneous, meaning that each instance belongs to the same class and entropy is one when the sample is equally divided between different classes.

$$E = -(1\log_2 1) = 0$$

$$E = -((0.5\log_2 0.5) + (0.5\log_2 0.5)) \\ = 1$$



INFORMATION GAIN

Information gain is precisely the measure used by ID3 to select the best attribute at each step in growing the tree.

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

1

Measure the effectiveness of an attribute in classifying a dataset.

2

We can use information gain to determine how good the splitting of nodes in a decision tree.

3

Information gain indicates how much information a particular feature/ variable give us about the final outcome.

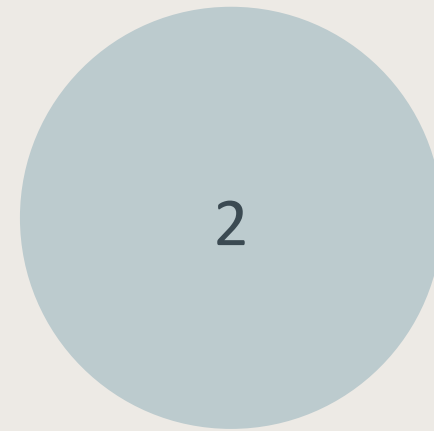


CONNECTION

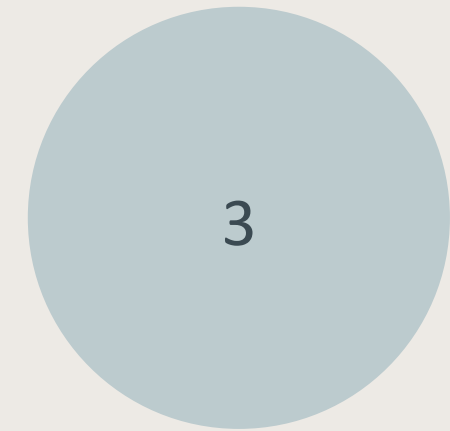
entropy and information gain



Information gain uses entropy to make decisions

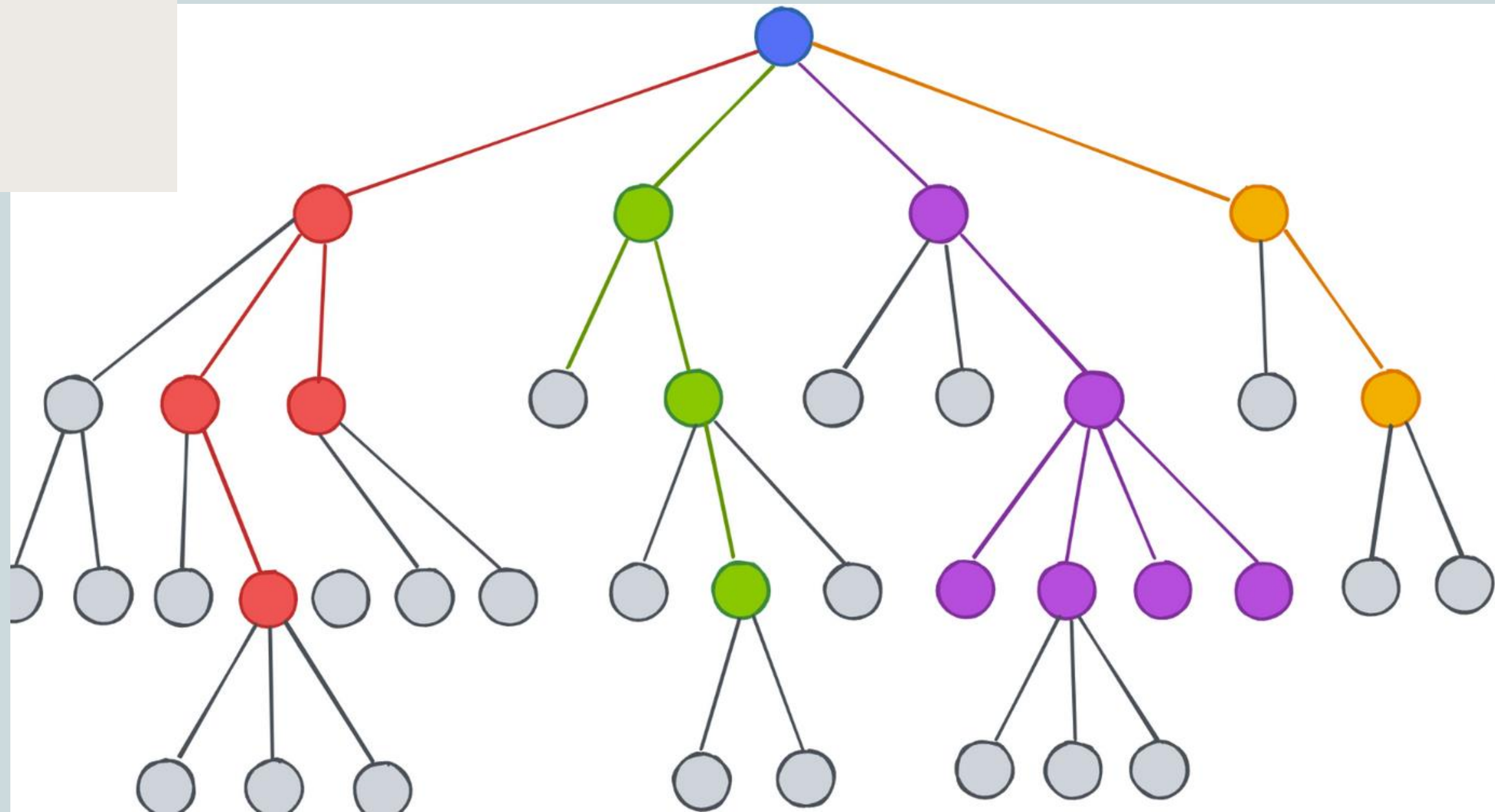


The information gain is based on the decrease in entropy after a dataset is split on an attribute.



The more the entropy removed, the greater the information gain. The higher the information gain, the better the split.

ID3 IMPLEMENTATION



FLOW CHART

